Cubic Function Report

Polynomial

$$f(q) = aq^3 + bq^2 + cq + d$$

Naive

$$g_{\text{naive}}(X) = X^3a + X^2b + Xc + d$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^3 + bq^2 + 2\frac{\Delta}{\epsilon}^2 (3aq + b) + cq + d$$

$$\begin{aligned}\mathrm{Var}(g_{\text{naive}}(X)) = 2\frac{\Delta}{\epsilon}^2 \Bigg(&360a^2\frac{\Delta}{\epsilon}^4 + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \\ &+ 2\frac{\Delta}{\epsilon}^2 \left(81a^2q^2 + 54abq + 12ac + 5b^2\right) + c^2\Bigg)\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{naive}}) = 2\frac{\Delta}{\epsilon}^2 \Bigg(&360a^2\frac{\Delta}{\epsilon}^4 + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \\ &+ 12\frac{\Delta}{\epsilon}^2 \left(15a^2q^2 + 10abq + 2ac + b^2\right) + c^2\Bigg)\end{aligned}$$

$$\mathrm{Bias}(g_{\text{naive}}) = 2\frac{\Delta}{\epsilon}^2 (3aq + b)$$

Unbiased

$$g_{\text{unb}}(X) = X^3a + X^2b + Xc - 2\frac{\Delta}{\epsilon}^2 (3Xa + b) + d$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3 + bq^2 + cq + d$$

$$\begin{aligned}\mathrm{Var}(g_{\text{unb}}(X)) = 2\frac{\Delta}{\epsilon}^2 \Bigg(&252a^2\frac{\Delta}{\epsilon}^4 + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \\ &+ 2\frac{\Delta}{\epsilon}^2 \left(63a^2q^2 + 42abq + 6ac + 5b^2\right) + c^2\Bigg)\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{unb}}) = 2\frac{\Delta}{\epsilon}^2 \Bigg(&252a^2\frac{\Delta}{\epsilon}^4 + 9a^2q^4 + 12abq^3 + 6acq^2 + 4b^2q^2 + 4bcq \\ &+ 2\frac{\Delta}{\epsilon}^2 \left(63a^2q^2 + 42abq + 6ac + 5b^2\right) + c^2\Bigg)\end{aligned}$$

Comparison

$$\text{mean gap} = -2\frac{\Delta}{\epsilon}^{2}(3aq+b)$$

$$\text{variance gap} = -24a\frac{\Delta}{\epsilon}^{4}\left(9a\frac{\Delta}{\epsilon}^{2} + 3aq^2 + 2bq + c\right)$$

$$\text{MSE gap} = -4\frac{\Delta}{\epsilon}^{4}\left(54a^2\frac{\Delta}{\epsilon}^{2} + 27a^2q^2 + 18abq + 6ac + b^2\right)$$

$$\text{Bias}(g_{\text{naive}})^2 = 4\frac{\Delta}{\epsilon}^{4}\left(9a^2q^2 + 6abq + b^2\right)$$